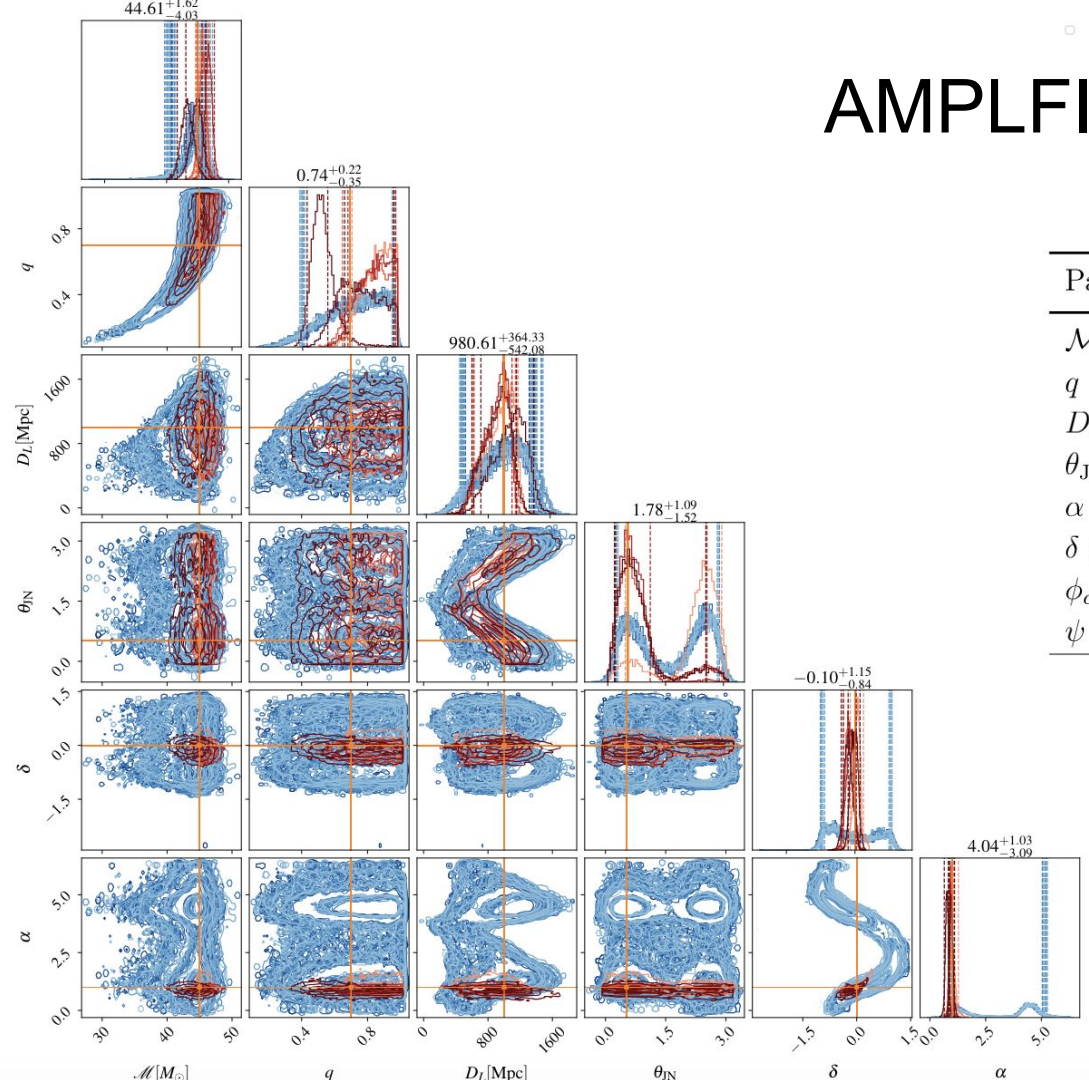
Optimizations to AMPLFI

Maanas Goel
Advisors: Dr. Deep Chatterjee, Dr. Erik Katsavounidis

AMPLFI



Parameter	Prior
$\mathcal{M}$	Uniform(10, 100)$M_\odot$
q	Uniform(0.125, 1)
D_L	Uniform in Vol.(100, 3000) Mpc ($\sim D_L^2$)
$\theta_{\rm JN}$	Sine(0, π)
α (RA)	Uniform(0, 2π)
δ (Dec.)	Cosine($-\pi/2$, $\pi/2$)
ϕ_c (Coal. phase)	Uniform(0, 2π)
ψ (Pol. angle)	Uniform(0, π)

Table 1: Prior distributions of parameters

Fig. 3: Posterior samples of AMPLFI (blue) vs Bibly (red)
https://arxiv.org/pdf/2407.19048

AMPLFI with splitting of parameters

Idea 1: 5-3 Split

- chirp_mass, mass_ratio, luminosity_ distance, phase, theta_jn
- ra, dec, psi

Idea 2: 2-6 Split

- chirp_mass, mass_ratio
- luminosity_distance, phase, theta_jn, ra, dec, psi

Gravitational wave polarizations:

$$h_+ = -\frac{4G\mu}{c^2 r}\left(\frac{v}{c}\right)^2 \cos 2\varphi$$

$$h_\times = -\frac{4G\mu}{c^2 r}\left(\frac{v}{c}\right)^2 \sin 2\varphi$$

Antenna pattern:

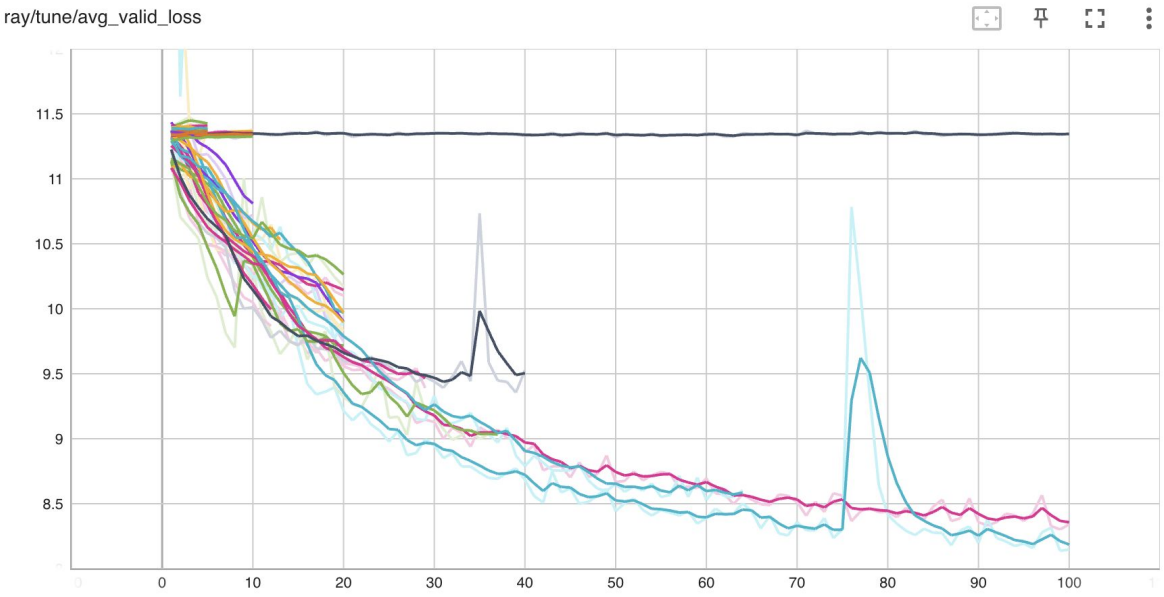
$F_{\times+}$(ra, dec, psi)

Detector response:

$$h_{\text{net}} = \mathbf{F}_\times h_\times + \mathbf{F}_+ h_+$$

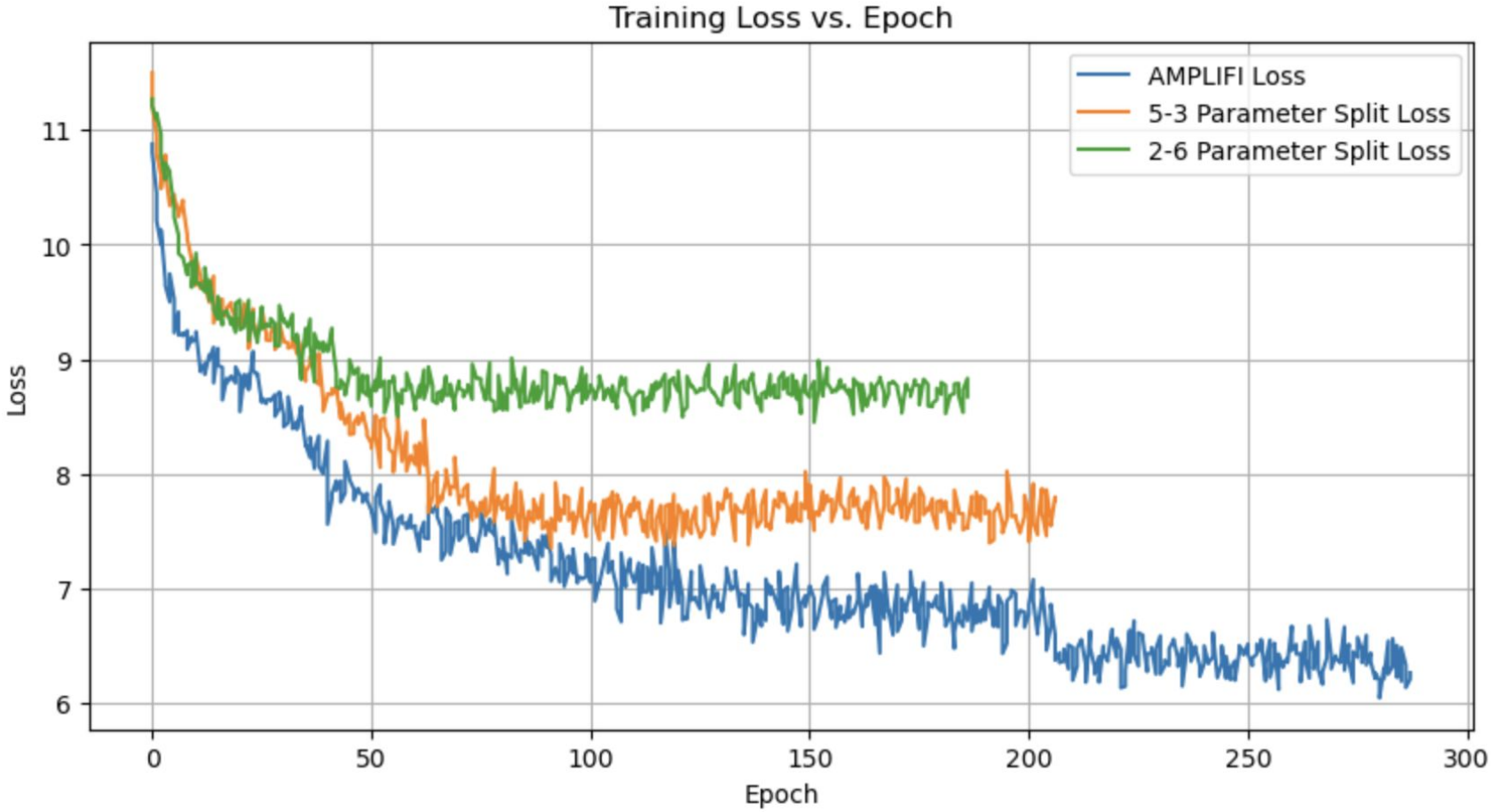
Creighton, J. D. E., & Anderson, W. G. (2011). Gravitational-Wave Physics and Astronomy: An Introduction to Theory and Experiment. Wiley-VCH.

Hyperparameter tuning for both splits




```
search_space = {
    "num_transforms2": tune.choice([40, 50, 60]),
    "num_blocks2": tune.choice([6, 8, 10]),
    "hidden_features2": tune.choice([100, 120, 150]),
    "num_transforms1": tune.choice([40, 50, 60]),
    "num_blocks1": tune.choice([6, 8, 10]),
    "hidden_features1": tune.choice([100, 120, 150]),
    "learning_rate1": tune.loguniform(1e-4, 1e-2),
    "learning_rate2": tune.loguniform(1e-4, 1e-2),
    "batch_size": tune.choice([300, 400, 200]),
    "momentum": tune.loguniform(1e-4, 1e-1),
    "weight_decay": tune.loguniform(1e-4, 1e-1),
    "optimizer_class": tune.choice([torch.optim.AdamW, torch.optim.SGD])
}
```




Training Loss vs. Epoch
AMPLIFI Loss
5-3 Parameter Split Loss
2-6 Parameter Split Loss
Loss
Epoch
11
10
9
8
7
6
0
50
100
150
200
250
300

Normalizing Flows on Tori and Spheres

https://arxiv.org/pdf/2002.02428

Using Normalizing Flows for Angular Parameters

- NCSF (Neural Circular Spline Flow) with flat base distribution
- Inference parameters ‘dec’ and ‘phi’ are periodic and angular parameters
- Transformations mapping 'dec' and 'phi' on a flat circular path, effectively creates a toroidal space from S^2 (two spheres)

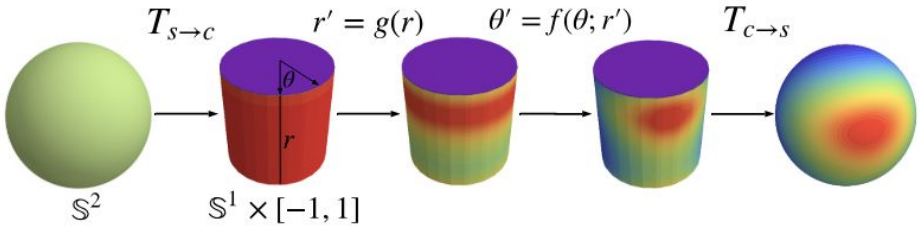
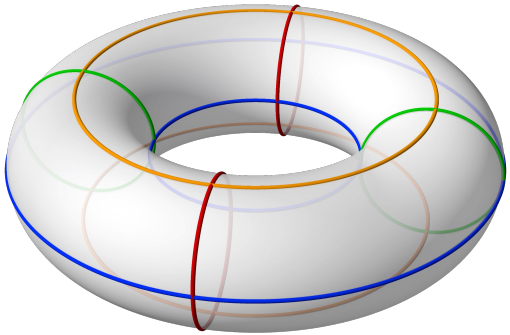
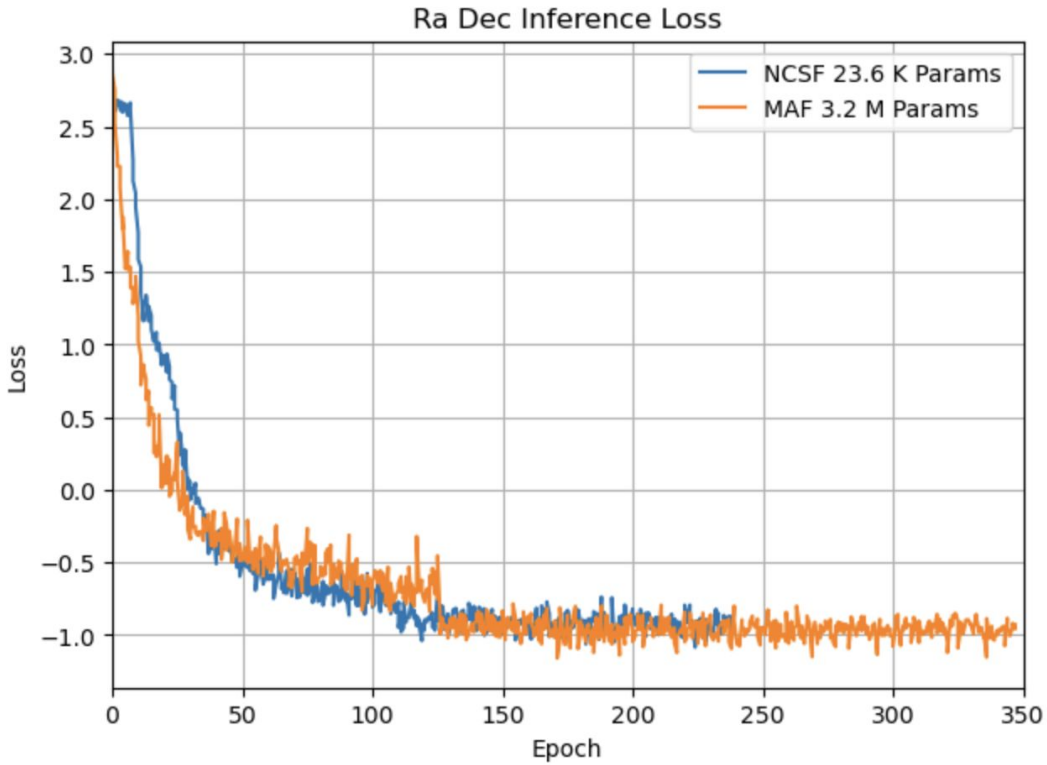


Figure 1. Illustration of the recursive flow on the sphere $\mathbb{S}^2$.



NCSF vs MAF

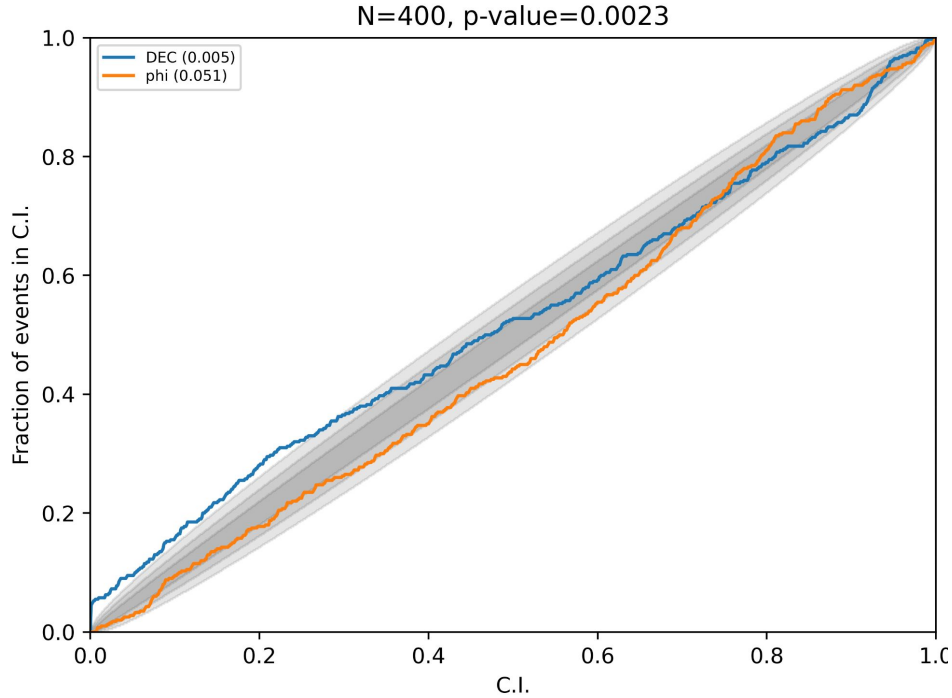
- MAF vs NCSF with only inferring on 2 parameters ‘dec’ and ‘ra’
- ‘dec’ ∈ (−π/2,π/2), ‘ra’ ∈ (0,2π)
- ‘luminosity_distance’: power law with injections distributed using the full luminosity distance prior (0.1 - 3 Gpc)
- Assigned fixed values for chirp_mass (50), mass_ratio (0.5), phase (π), theta_jn (π/6), psi (π/2)

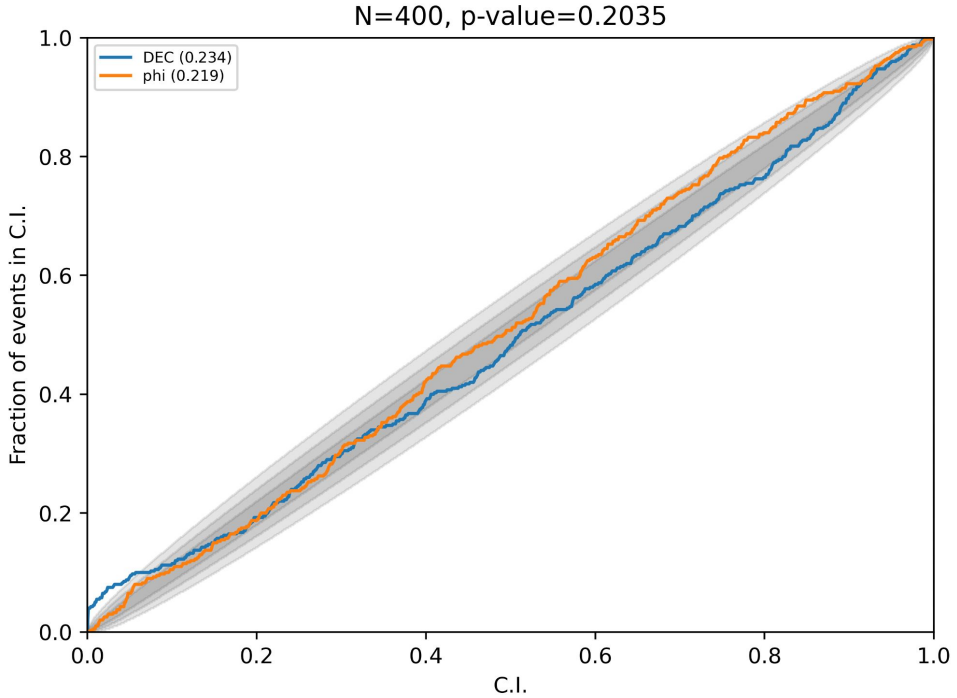


NCSF training time ≈ 4 hrs
MAF training time ≈ 11 hrs

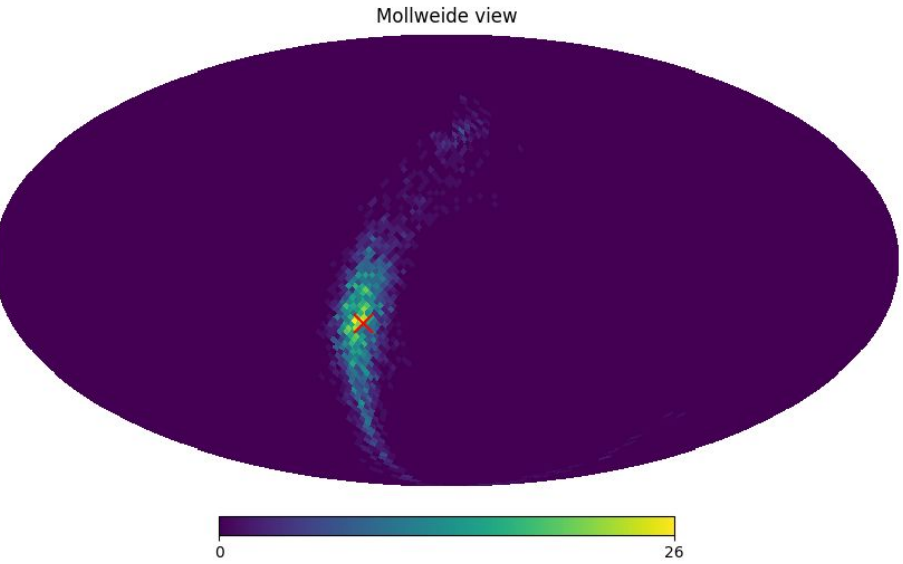
PP Plots

MAF vs NCSF

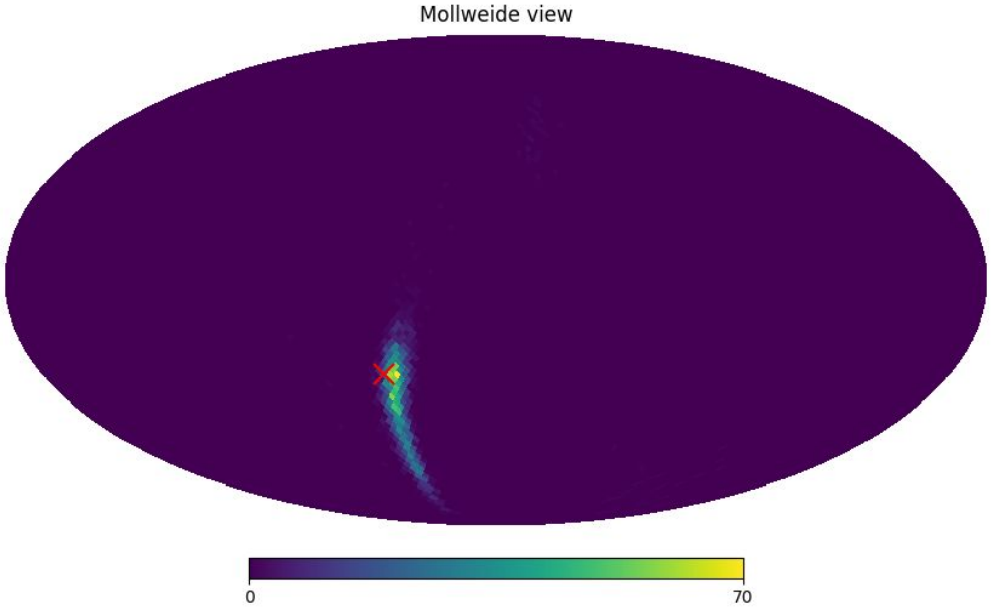




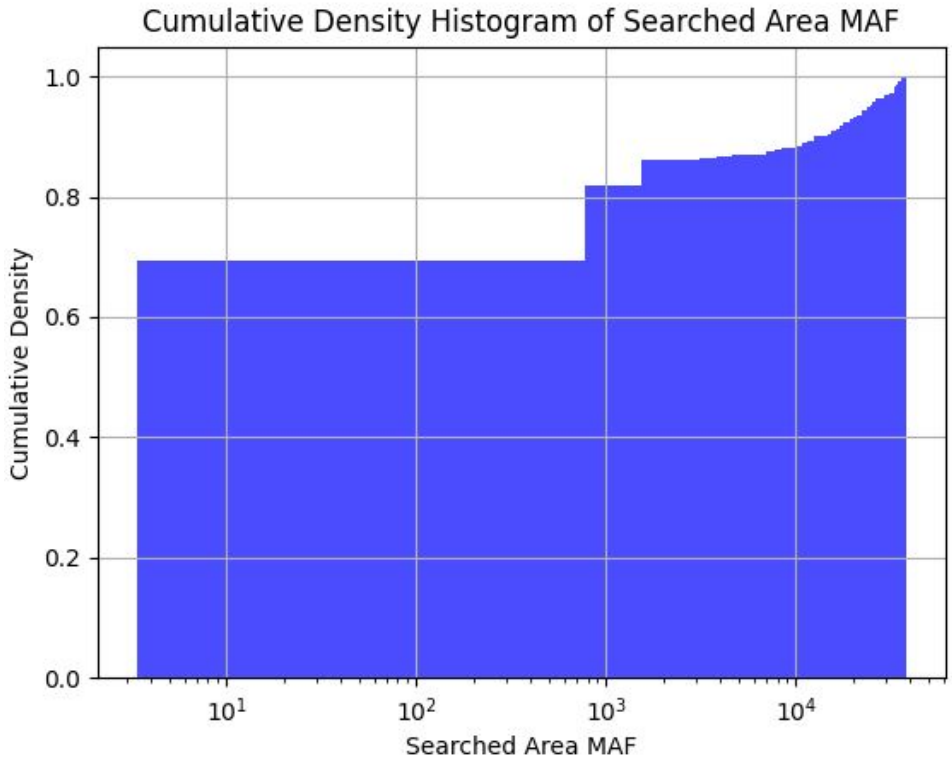
MAF

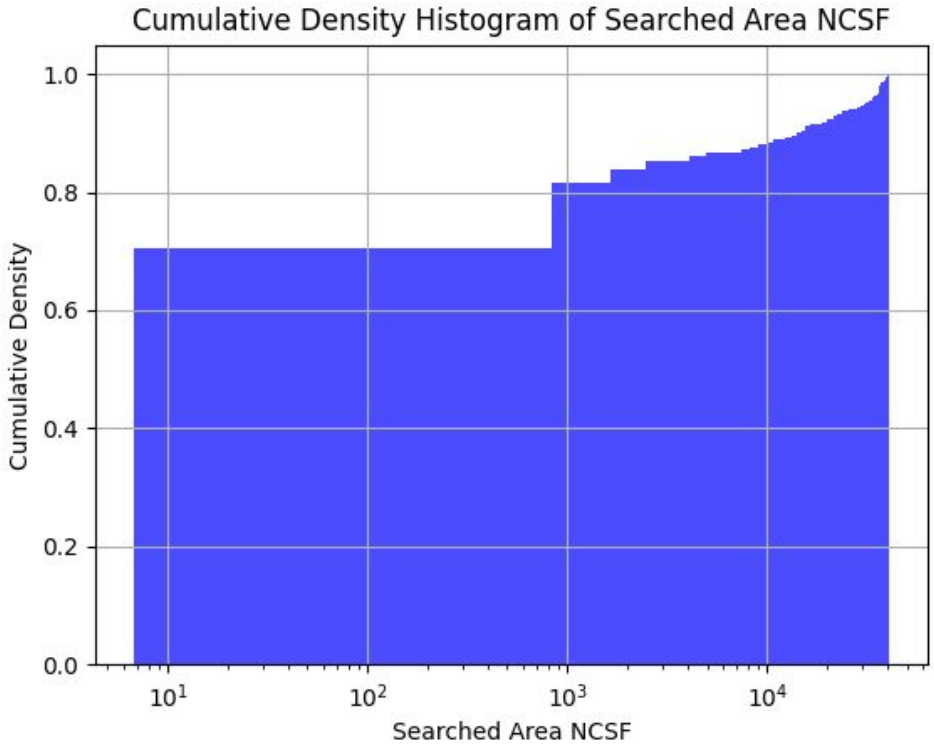


NCSF



Searched Area

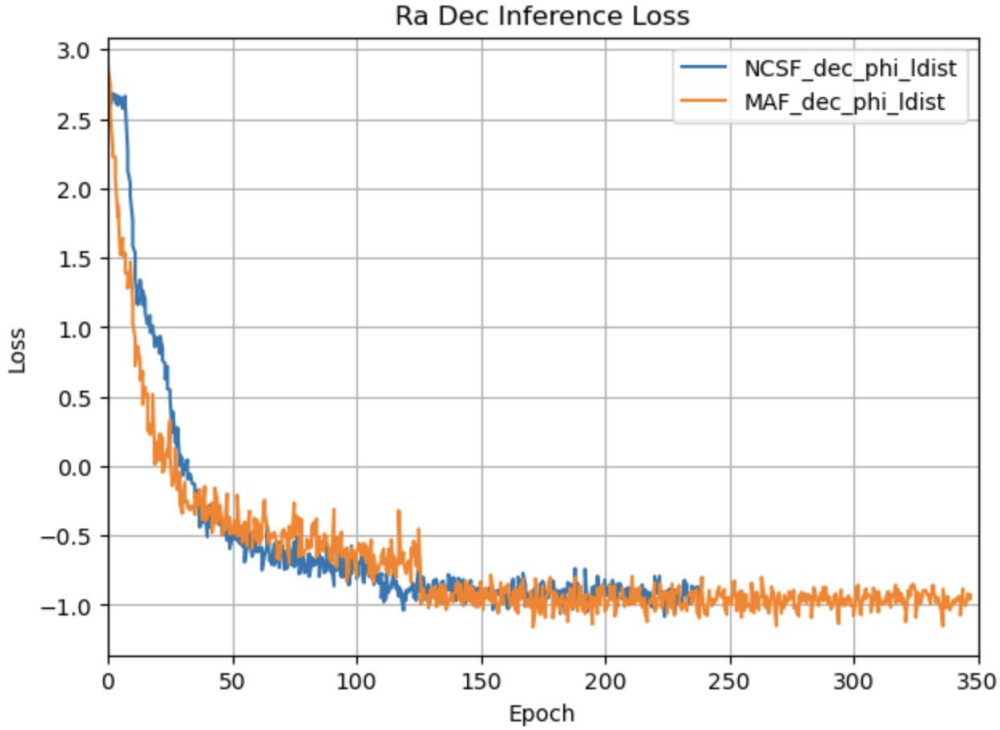




Searched Area Comparisons

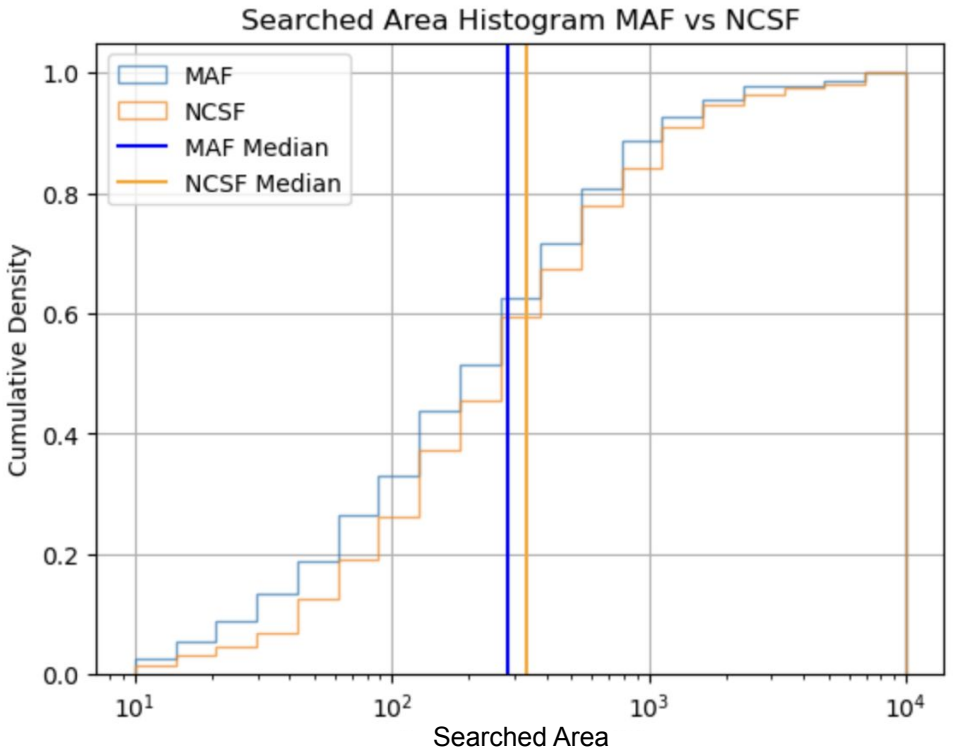
Training (Infer on ra,dec) (Full LD prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2



Testing (Full LD prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2

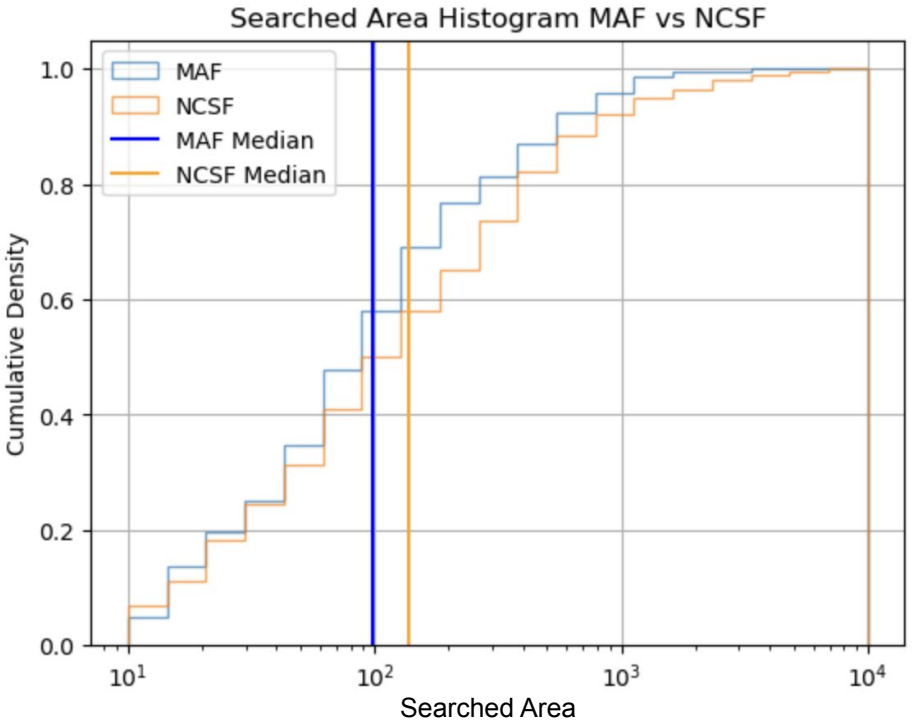



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(282.00266479095205, 332.36028350362204)
```

Testing (Fixed LD prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 1 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2

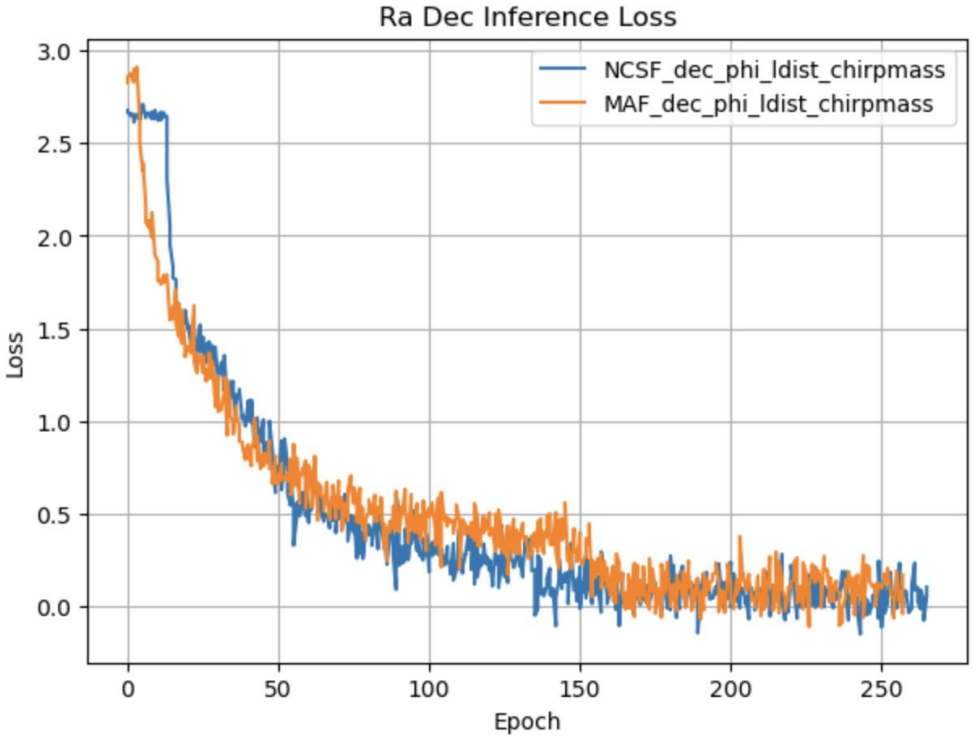



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(97.35806284449535, 137.64415781463134)
```

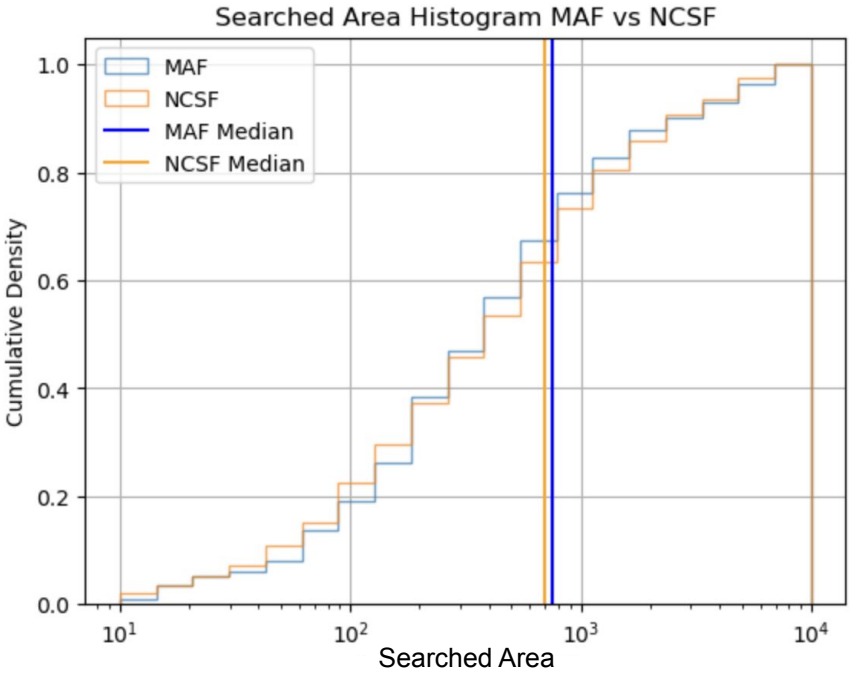
Training (Infer on ra,dec) (Full LD, Chirp Mass prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 10 - 100
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2



Testing (Full LD, Chirp Mass prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 10 - 100
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2

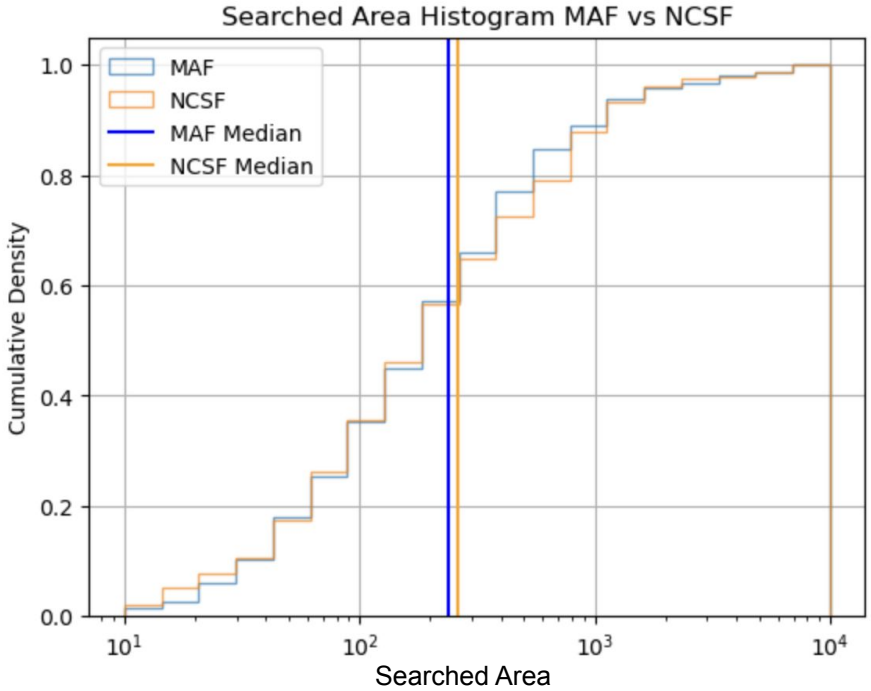



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(750.3285188187831, 699.970900106113)
```

Testing (Fixed LD, Chirp Mass prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 1 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2

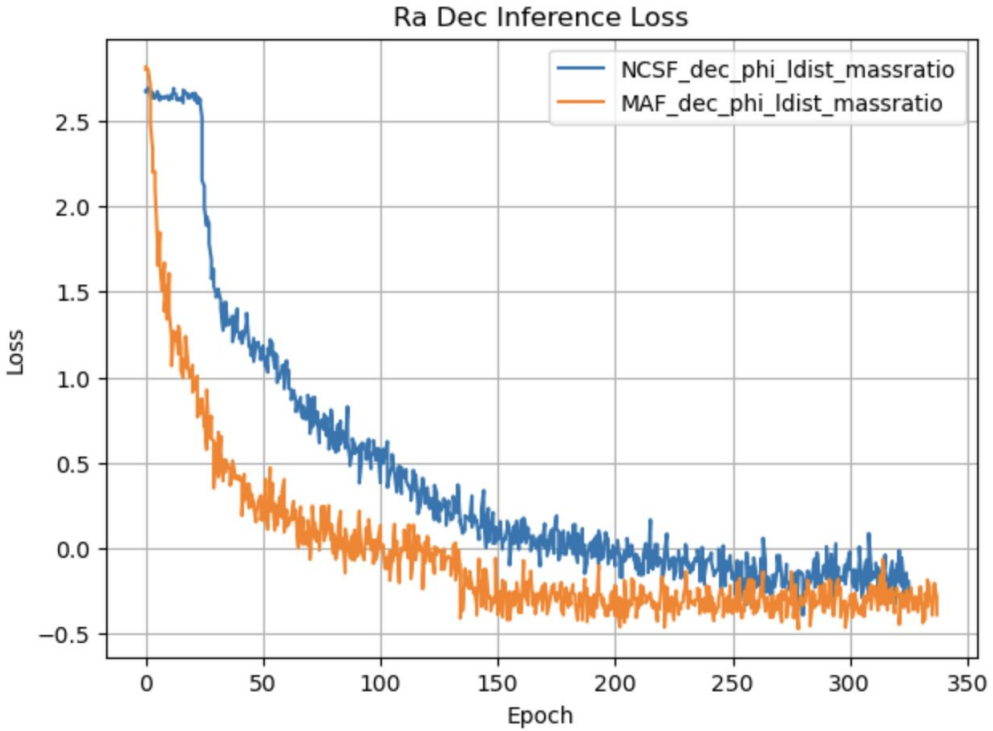



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(238.35939523997138, 261.85961730588406)
```

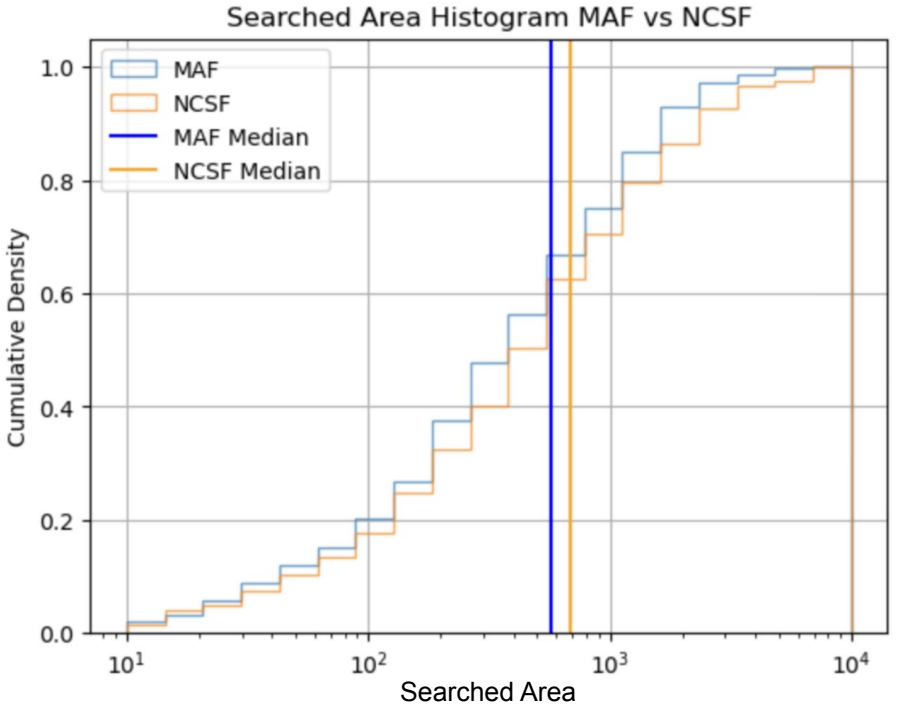
Training (Infer on ra,dec) (Full LD, Mass Ratio prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.125 - 0.999
- phase: π
- theta_jn: π/6
- psi: π/2



Testing (Full LD, Mass Ratio prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.125 - 0.999
- phase: π
- theta_jn: π/6
- psi: π/2

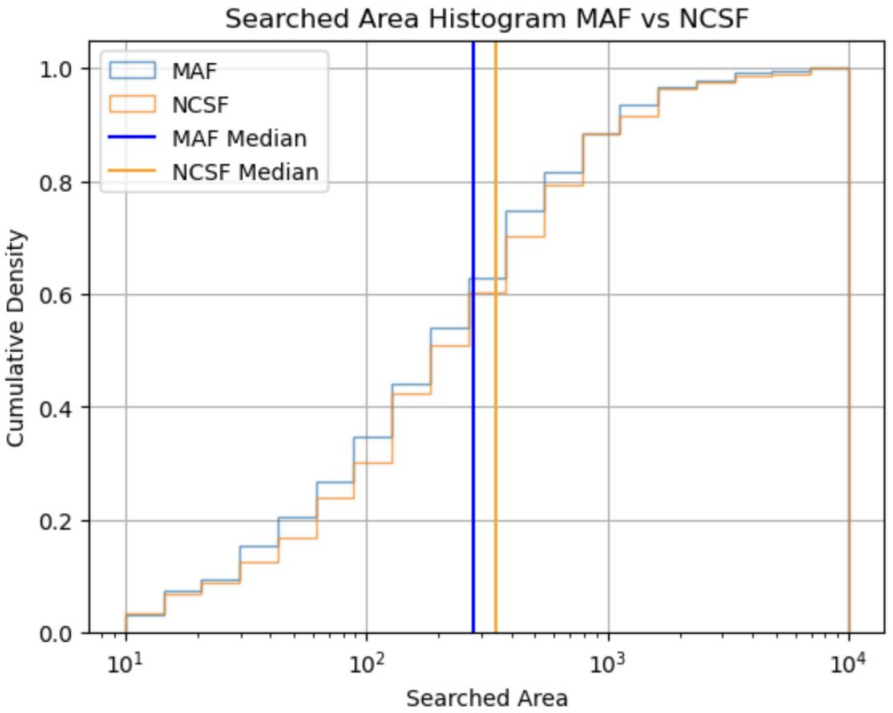



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(570.7196787435935, 691.5779636540015)
```

Testing (Fixed LD, Mass Ratio prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 1 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: π/2

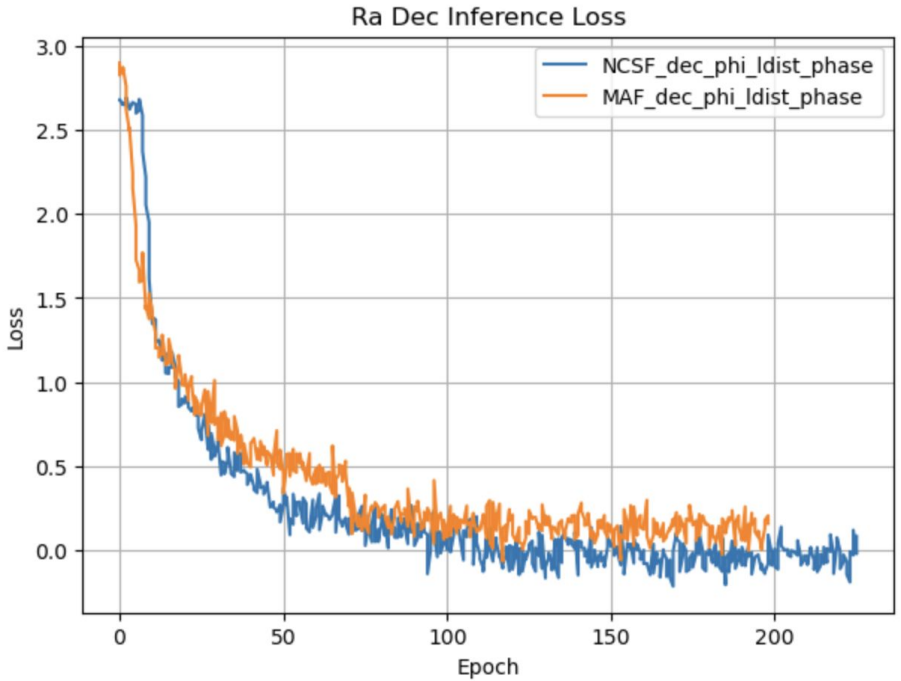



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(278.64549021010737, 344.1103945365784)
```

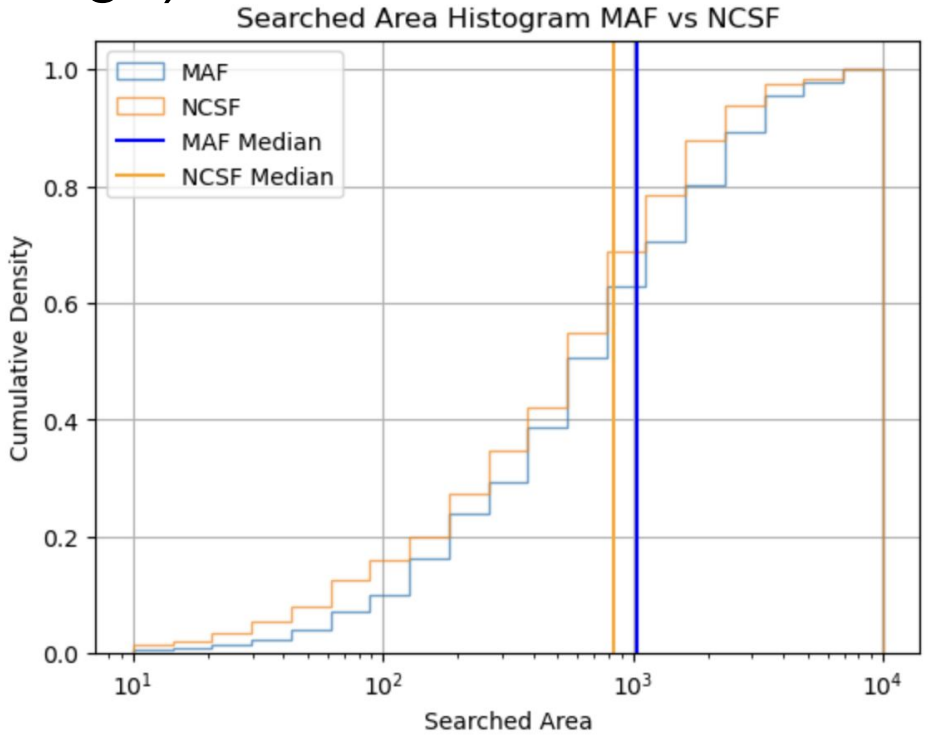
Training (Infer on ra,dec) (Full LD, Phase prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: 0 - 2π
- theta_jn: π/6
- psi: π/2



Testing (Full LD, Phase prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: 0 - 2π
- theta_jn: π/6
- psi: π/2

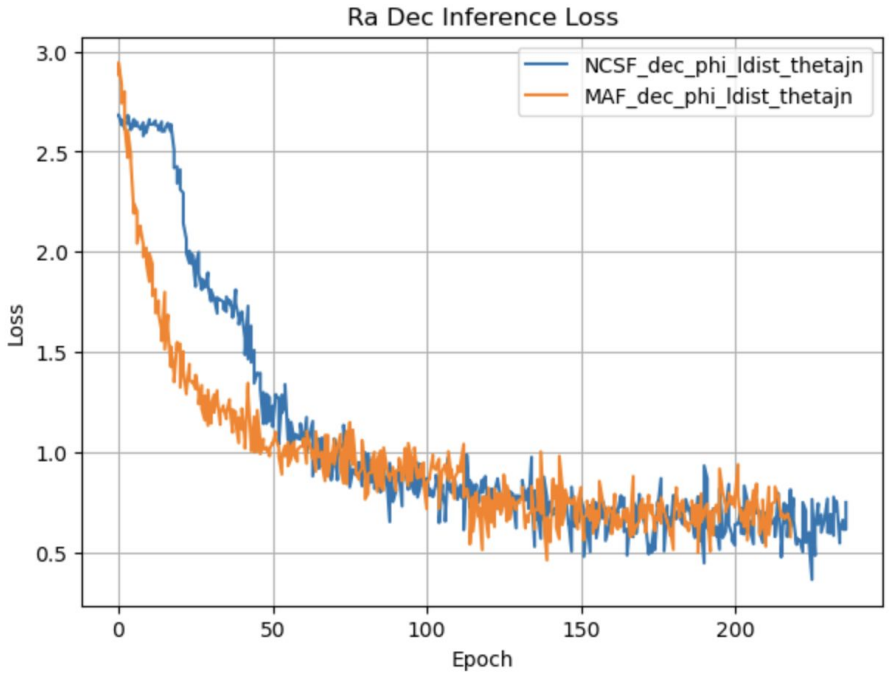



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(1045.7598819331138, 837.6150579207444)
```

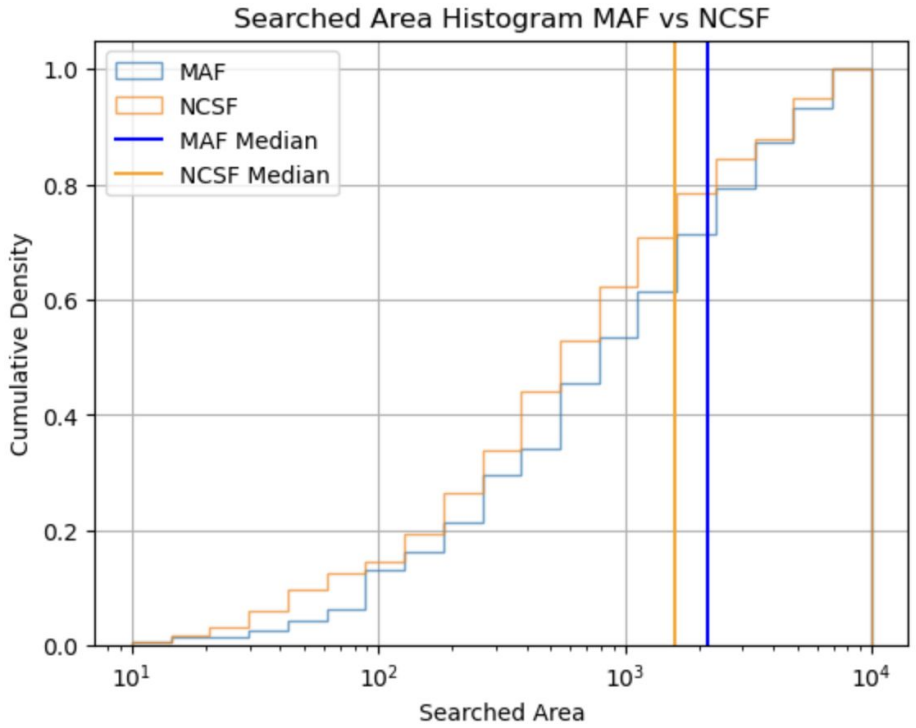
Training (Infer on ra,dec) (Full LD, Theta Jn prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: 0 - π
- psi: π/2



Testing (Full LD, Theta Jn prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: 0 - π
- psi: π/2

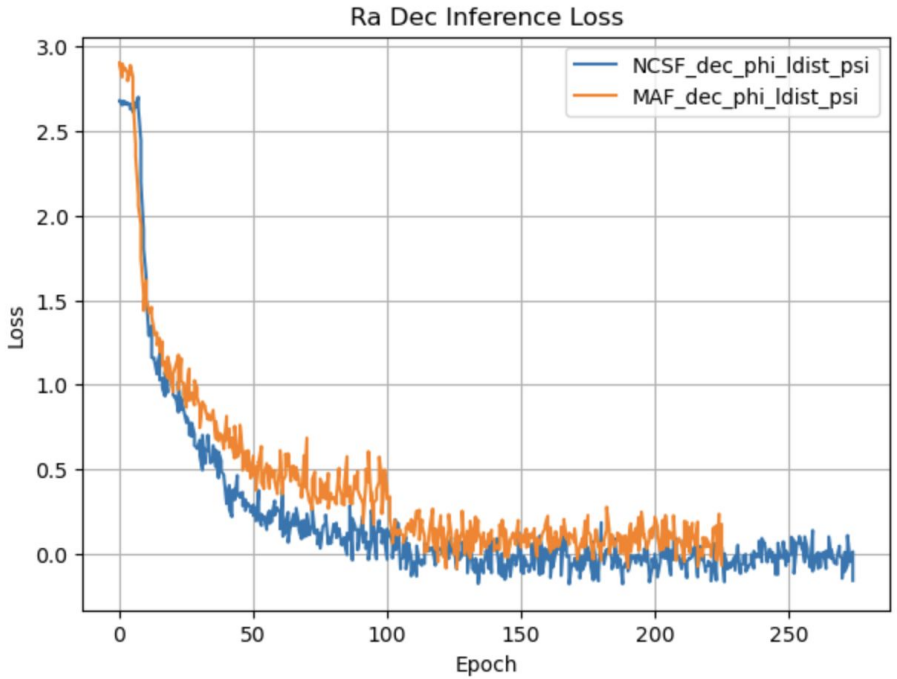



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(2170.413366516077, 1596.3365131916394)
```

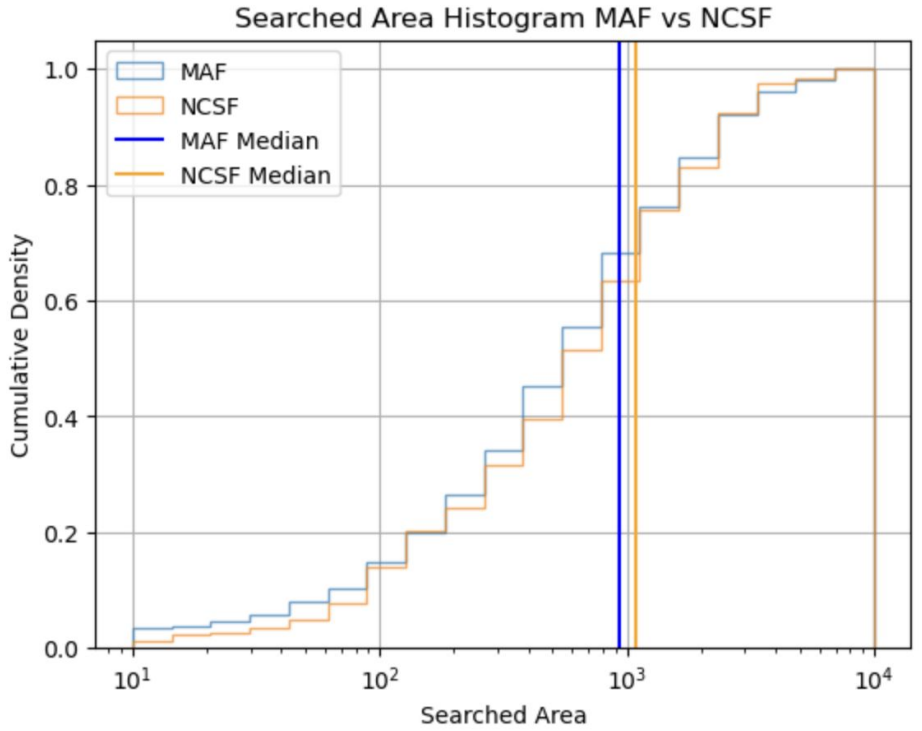
Training (Infer on ra,dec) (Full LD, Psi prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: 0 - π



Testing (Full LD, Psi prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: π/6
- psi: 0 - π

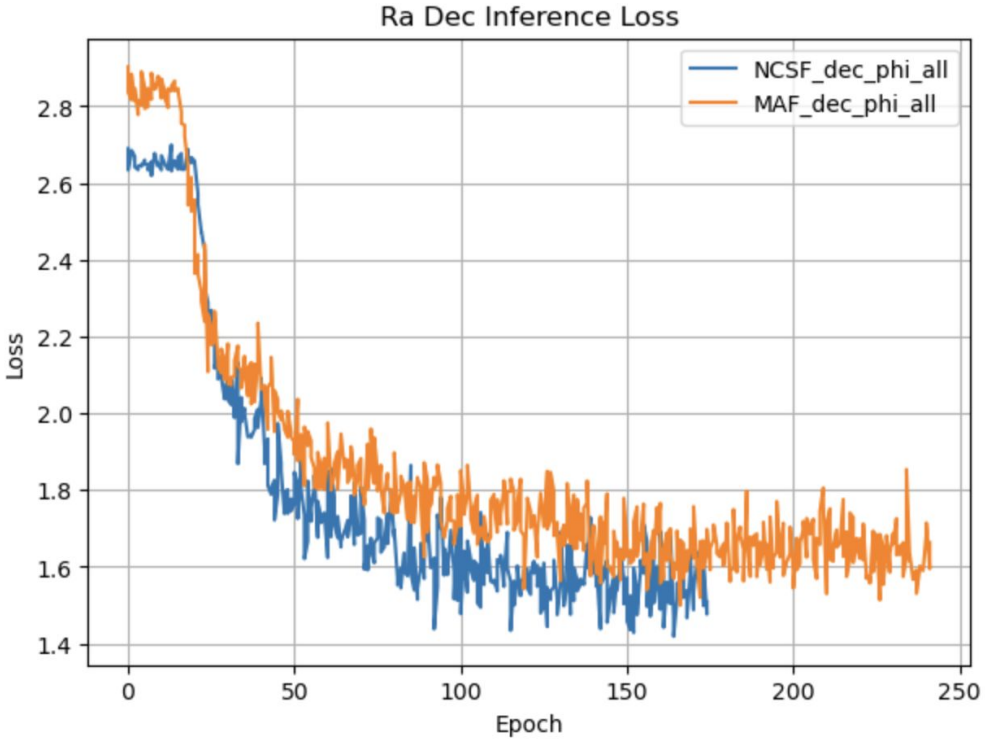



```
searched_areas_maf_median, searched_areas_ncsf_median
```

```
(936.6517080556621, 1089.4031514840945)
```

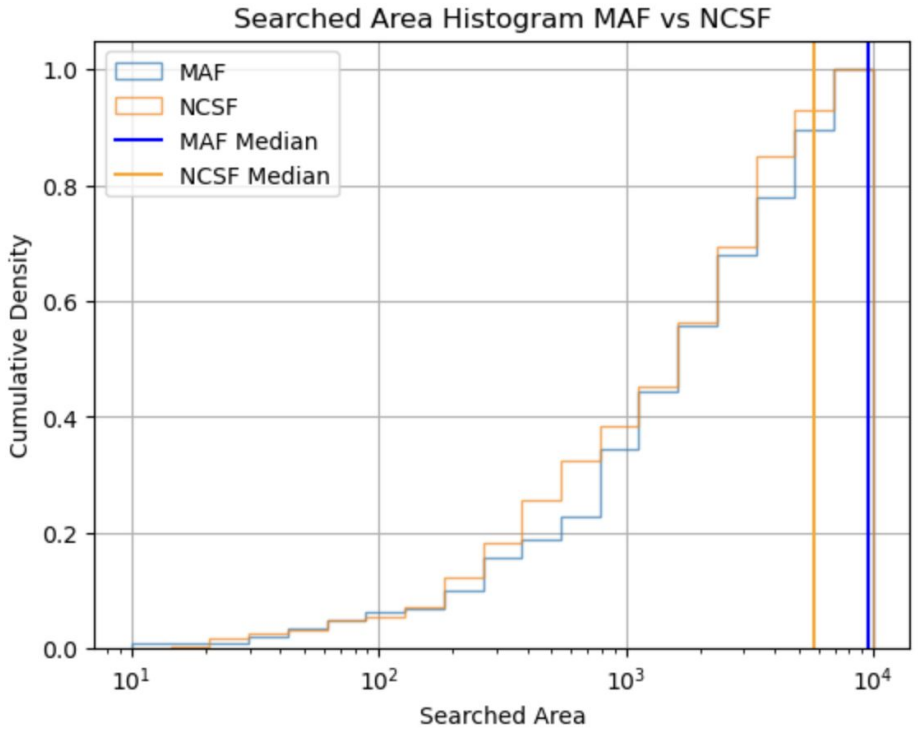
Training (Infer on ra,dec) (All parameters full prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 10 - 100
- mass_ratio: 0.125 - 0.999
- phase: 0 - 2π
- theta_jn: 0 - π
- psi: 0 - π



Testing (All parameters full prior range)

- dec ∈ (−π/2,π/2)
- ra ∈ (0,2π)
- luminosity_distance: 0.1 - 3 Gpc
- chirp_mass: 10 - 100
- mass_ratio: 0.125 - 0.999
- phase: 0 - 2π
- theta_jn: 0 - π
- psi: 0 - π



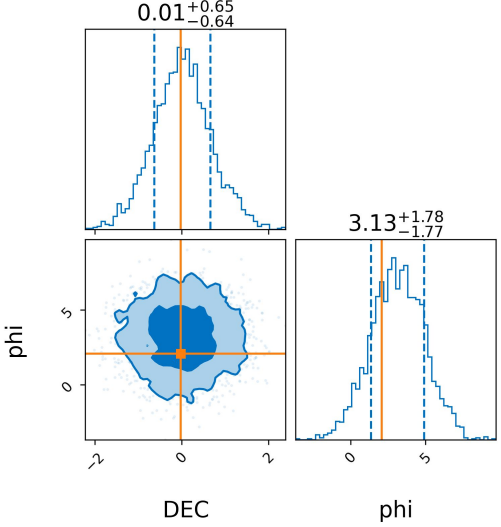

```
searched_areas_maf_median, searched_areas_ncsf_median
```

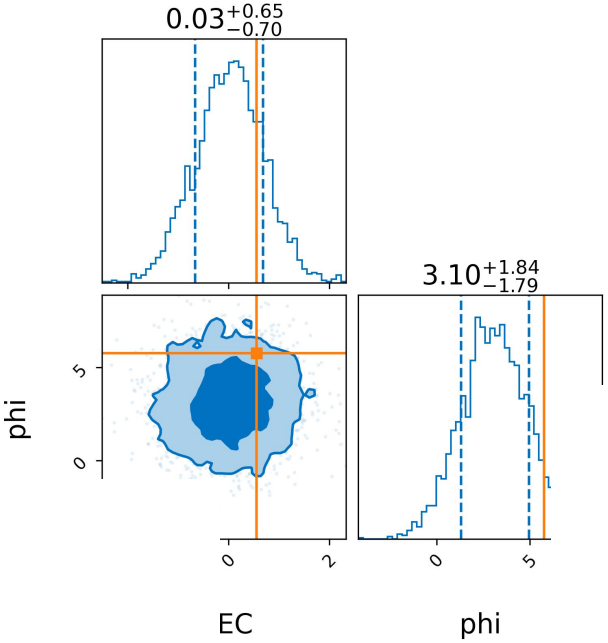
```
(9571.304729988147, 5794.483326537896)
```

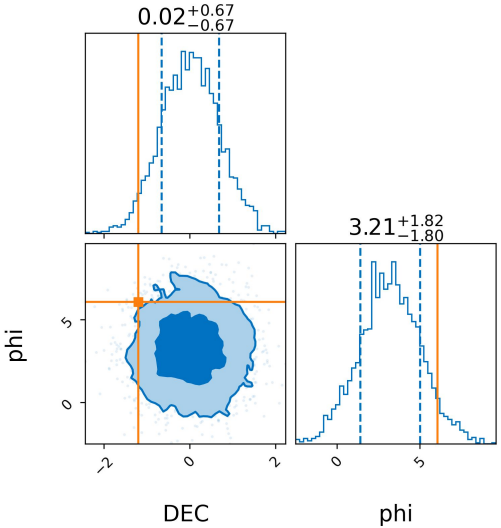
Future Work

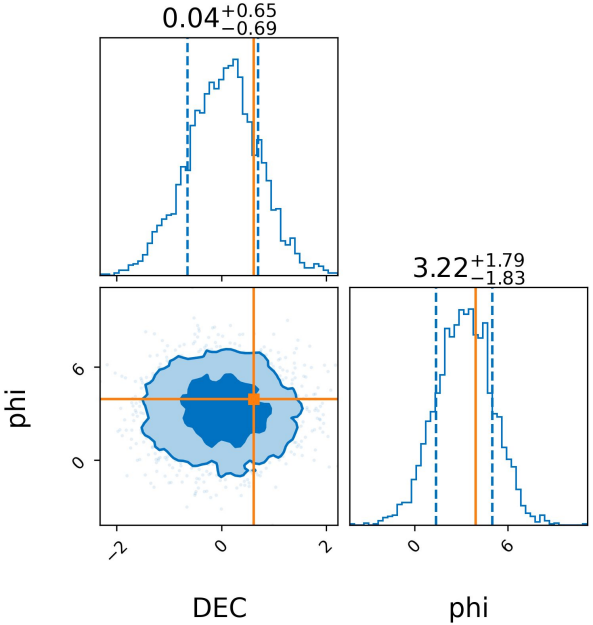
- NCSF is a promising approach
- Compare all 8 parameters with the NCSF flow and MAF flow
- Incorporate into AMPLFI codebase https://github.com/ML4GW/amplfi

MAF Cornerplots

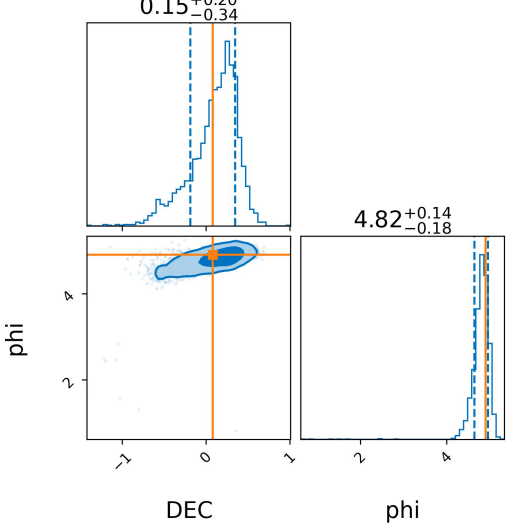


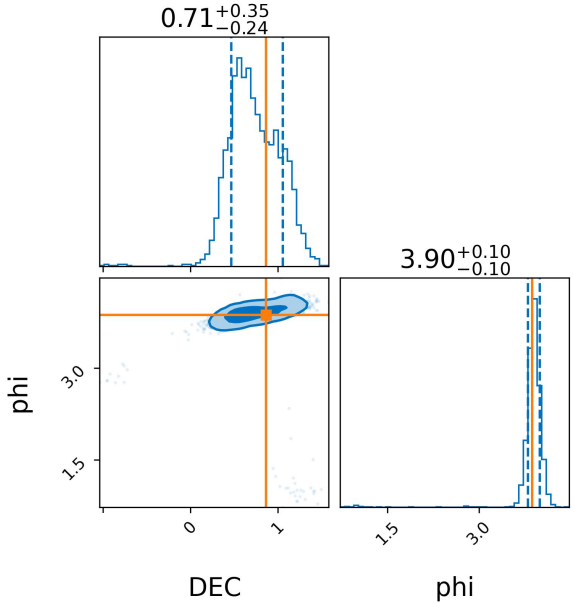


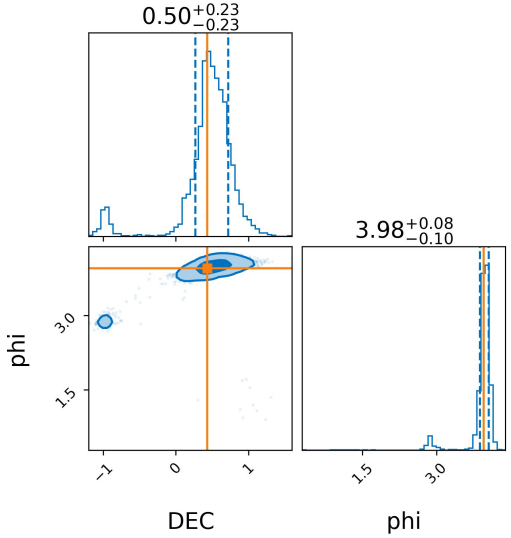


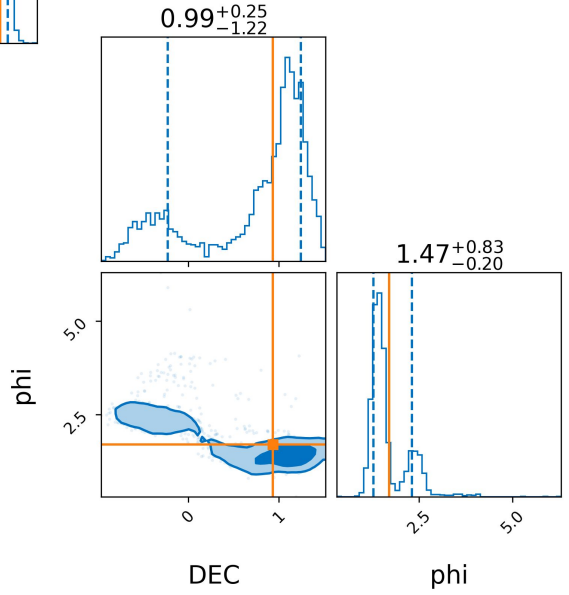


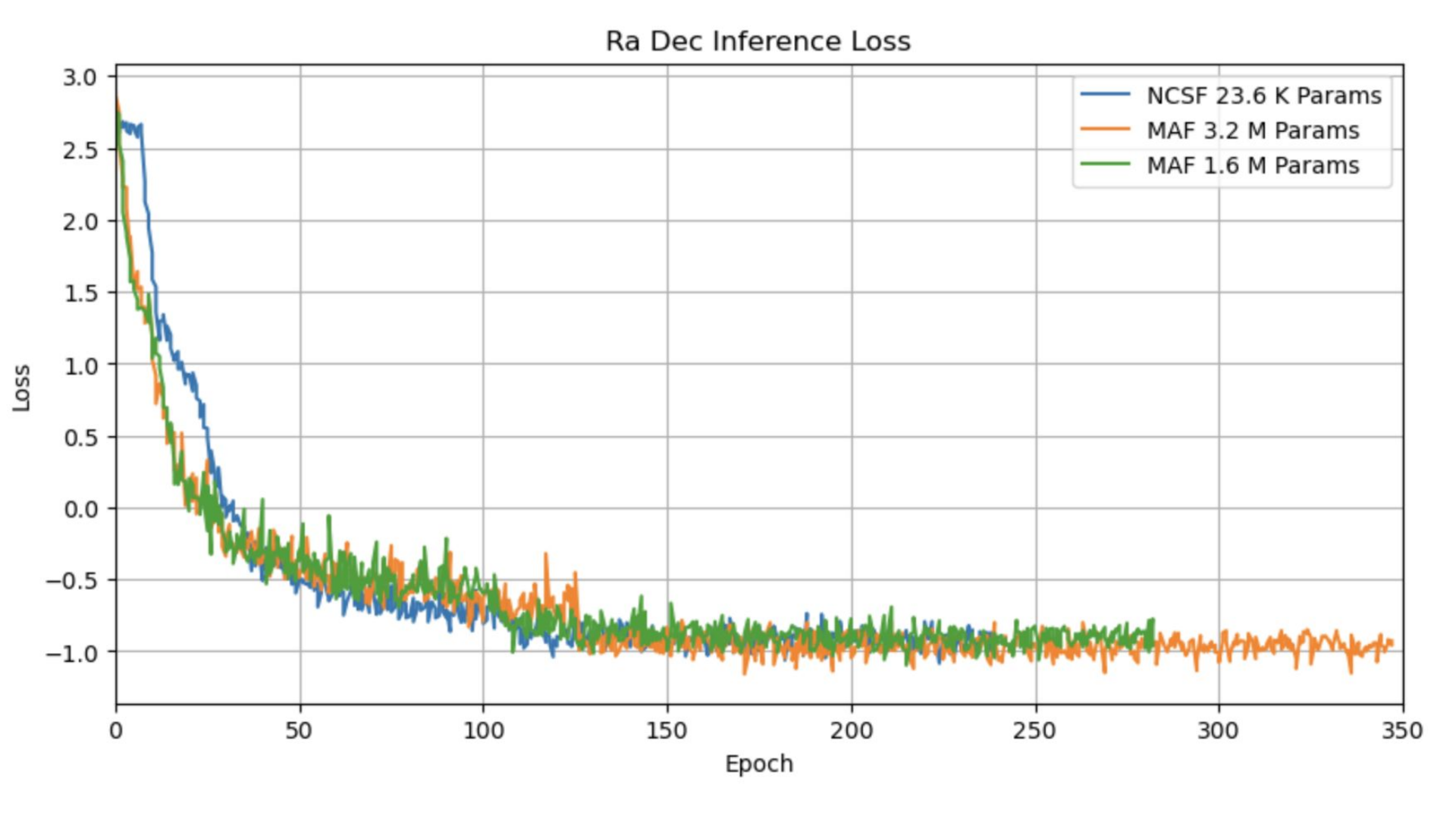
NCSF Cornerplots










Ra Dec Inference Loss
NCSF 23.6 K Params
MAF 3.2 M Params
MAF 1.6 M Params
Loss
Epoch
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
50
100
150
200
250
300
350

Comparison

AMPLIFI Parameters

	Name	Type	Params
0	embedding_net	ResNet	2.6 M
1	transforms	ConditionalComposeTransformModule	3.2 M

Total Parameters
= 5.8 M

NCSF Parameters

	Name	Type	Params
0	embedding_net	ResNet	2.6 M
1	NCSF_flow	NCSF	23.6 K

Total Parameters
= 2.6 M

Testing (Fixed LD, Phase prior range)

- dec $\in (-\pi/2, \pi/2)$
- ra $\in (0, 2\pi)$
- luminosity_distance: 1 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: $\pi/6$
- psi: $\pi/2$

Testing (Fixed LD, Theta Jn prior range)

- dec $\in (-\pi/2, \pi/2)$
- ra $\in (0, 2\pi)$
- luminosity_distance: 1 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: $\pi/6$
- psi: $\pi/2$

Testing (Fixed LD, Psi prior range)

- dec $\in (-\pi/2, \pi/2)$
- ra $\in (0, 2\pi)$
- luminosity_distance: 1 Gpc
- chirp_mass: 50
- mass_ratio: 0.5
- phase: π
- theta_jn: $\pi/6$
- psi: $\pi/2$